

Checkpointed Newton–Nesterov (CN^2): A Stage-Wise Switching Framework for Hybrid First– and Second–Order Optimization

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Abstract

We introduce a stage-wise switching framework, *Checkpointed Newton–Nesterov* (CN^2), that alternates between a coarse *Nesterov accelerated gradient* (NAG) phase and a local *Newton* phase. Rather than continuously blending momentum with curvature information—as in recent inertial cubic-regularized Newton methods— CN^2 performs a *hard* switch: when the Newton decrement $\lambda(x) = \sqrt{\nabla f(x)^\top [H(x)]^{-1} \nabla f(x)}$ indicates we have left the solution basin, we navigate with cheap first-order accelerated steps; once the decrement falls below a user threshold τ , we commit to damped Newton for quadratic local convergence. In nonconvex regions where the Hessian is not positive definite, we fall back to a gradient-norm gate, making the method robust on functions such as Rosenbrock and Beale where vanilla Newton fails abruptly. On a battery of test problems, CN^2 converges on every instance—including Rosenbrock in dimension $n = 200$, where plain Newton, Newton-CG, and L-BFGS all stall—while using a comparable or smaller number of Hessian evaluations. The method is robust to the user parameter K_0 (jump length) and exposes a clean accuracy-cost trade-off in the threshold τ .

1 Introduction

Newton’s method attains quadratic local convergence but incurs an $\mathcal{O}(n^3)$ cost per iteration for the dense Hessian solve, rendering it prohibitive at scale. Nesterov accelerated gradient (NAG) methods achieve the optimal $\mathcal{O}(1/k^2)$ rate for smooth convex problems but slow down geometrically near the solution, especially on ill-conditioned objectives. This tension motivates hybrid designs that recover higher-order local behavior without paying the full second-order cost on every step.

Recent work has pursued *continuous* blending of momentum with cubic regularization, *e.g.* inertial cubic-regularized Newton (ICN) [?], accelerated cubic Newton variants [?], and momentum-accelerated stochastic cubic methods [?]. These methods intertwine momentum and curvature on a per-step basis. In contrast, this paper studies a sharper, arguably more interpretable design: we keep the two algorithmic families *separate* and switch between them only at coarse *checkpoints*. This is the “coarse-to-fine navigation” philosophy: expensive Newton assessments are performed rarely, and accelerated gradient steps fill the gaps.

Our contribution is a principled *stage-wise* method that:

1. runs a short Newton burst (two damped steps) and evaluates the Newton decrement;
2. if the decrement exceeds a threshold (far from the solution), runs a Nesterov phase until the decrement—measured with a rare, dimension-adaptive Hessian cadence—enters the solution basin;

3. falls back to a gradient-norm gate in nonconvex regions where the Hessian is not positive definite, ensuring global robustness.

2 Background and Related Work

Newton and trust regions. Damped Newton with backtracking retains global convergence on convex objectives subject to Lipschitz-continuous Hessians. Newton–CG [?] avoids forming H by solving the Newton system with conjugate gradients. Quasi-Newton methods such as L-BFGS approximate curvature from gradient differences and need no Hessian at all.

Nesterov acceleration. The accelerated gradient method [?] reaches $f(x_k) - f^* = \mathcal{O}(1/k^2)$ for L -smooth convex functions. It is attractive when evaluations of the gradient are cheap relative to second-order information.

Cubic regularization. Trust-region and cubic-regularized Newton (CRN) [?, ?] restore global convergence while achieving $\mathcal{O}(1/k^{3/2})$ (convex) and superlinear/quadratic rates near the solution, at the price of a subproblem solve involving the Hessian.

Recent hybridizations (2025–2026). Inertial cubic-regularized Newton (ICN) [?] explicitly fuses NAG-style momentum with the cubic framework; OPTAMI/NATA [?] provide accelerated tensor methods; and momentum-accelerated stochastic cubic variants [?] tackle the stochastic regime. Our work differs by keeping the phases *disjoint* and switching on a *geometric* measure (the Newton decrement), rather than blending operators.

3 The CN^2 Algorithm

3.1 Setup and notation

We minimize a smooth (possibly nonconvex) function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Let ∇f denote the gradient and $H = \nabla^2 f$ the Hessian. The *Newton decrement* is

$$\lambda(x) = \sqrt{\nabla f(x)^\top (H(x) + \rho I)^{-1} \nabla f(x)}, \quad (1)$$

where $\rho \geq 0$ is a small regularization shift. For convex objectives with $H \succ 0$, $\lambda(x)$ is a natural proximity measure to the solution.

3.2 Dual entry gate

When H is not positive definite (nonconvex region), the decrement (1) is not meaningful. We therefore define the entry measure

$$g(x) = \begin{cases} \lambda(x), & H(x) \succ 0, \\ \|\nabla f(x)\|, & \text{otherwise,} \end{cases} \quad (2)$$

and compare it against a threshold τ (for the Hessian case) or τ_g (for the gradient case). This dual gate (option “c”) is what makes the method robust on nonconvex test functions such as Rosenbrock and Beale.

Algorithm 1 CN^2 (Checkpointed Newton–Nesterov)

- 1: Given x_0 , thresholds $\tau, \tau_g > 0$, jump length K_0 , Newton steps $m = 2$.
 - 2: **repeat**
 - 3: **Newton burst:** run m damped-Newton steps, each with a fresh Hessian.
 - 4: **Evaluate:** compute the entry measure $g(x)$ from a fresh Hessian.
 - 5: **if** $g(x) \leq \text{threshold}$ **then**
 - 6: **return** converged at x .
 - 7: **end if**
 - 8: **NAG phase:** run K_0 Nesterov steps with backtracking step size, re-checking $g(x)$ every $K_{ck} = \max(20, \lfloor n/5 \rfloor)$ steps (fresh Hessian).
 - 9: **until** the entry measure falls below the threshold
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The check cadence K_{ck} grows with dimension: for larger n , Hessian evaluations become relatively more expensive, so we probe $g(x)$ less often. Between probes we reuse the last Hessian (option “ii”), so only a few Hessians are ever formed.

4 Convergence

We state convergence conditions for CN^2 . We work with the following standard assumptions.

Assumption 1 (Smoothness). *f is twice differentiable with L -Lipschitz gradient, and its Hessian satisfies $\|H(x) - H(y)\| \leq M\|x - y\|$ on the sublevel set $\{x : f(x) \leq f(x_0)\}$.*

Assumption 2 (Injective solution basin). *f is μ -strongly convex in the “solution basin” $\Omega = \{x : \lambda(x) \leq \tau\}$, so $H(x) \succeq \mu I$ for $x \in \Omega$.*

Theorem 1 (Termination in the solution basin). *Let ?? hold. If a Nesterov phase converges (in the sense that successive gradients vanish) or the Hessian cadence eventually probes a point x with entry measure $g(x) \leq \tau$ (resp. τ_g), then ?? enters Ω and switches to the Newton phase.*

Proof. (Sketch.) Under ??, a NAG step with backtracking is a descent step that lower-bounds the decrease of f by $\frac{1}{2}\alpha\|\nabla f\|^2$. Hence, if the NAG phase is run long enough without an early exit, the gradient norm decays and eventually the gradient gate $g(x) = \|\nabla f\| \leq \tau_g$ is triggered (or a probed Hessian reveals $\lambda \leq \tau$). At that moment we are in Ω and the Newton phase takes over. $\square$

Theorem 2 (Local quadratic convergence). *Let ???? hold and suppose the NAG phase has delivered $x \in \Omega$ with $\lambda(x) \leq \tau$. Then the damped-Newton phase converges quadratically to the unique minimizer x^* : there is a constant $C > 0$ such that*

$$\|x_{k+1} - x^*\| \leq C \|x_k - x^*\|^2$$

once the damping factor reaches 1.

Proof. (Sketch.) Within Ω the Hessian is uniformly positive definite (??), so the pure Newton step is a descent direction and the backtracking accepts step size 1. Standard Newton theory under ?? then gives local quadratic convergence with constant $C \sim M/(2\mu)$. $\square$

We next study the checkpoint subsequence $\{x_{jK}\}$, i.e. the iterates at the start of each cycle’s Newton phase, and the two-phase decay of the entry measure $\delta(x)$, which the dual gate sets to $\lambda(x)$ in Ω and to $\|\nabla f(x)\|$ outside it (Sec. ??).

Lemma 1 (Descent blocks). *Under ??, every Nesterov block (Lemma 1 style backtracking) and every damped-Newton block is a descent sequence, and the accepted steps remove at least $\frac{1}{2}\alpha\|\nabla f\|^2$ of suboptimality.*

Lemma 2 (Linear near-field decay). *Under ????, inside Ω the λ -gate applies and Nesterov AGD for a μ -strongly convex, L -smooth f contracts the entry measure linearly: $\lambda(x_{:k}) \leq c\rho^k$ with $\rho = 1 - \sqrt{\mu/L}$. That is, $p \simeq 1$ in the near field.*

Lemma 3 (Far-field superlinear transit). *If along a Nesterov block the entry measure obeys a log-convex contraction $\log \delta_{k+1} = p_k \log \delta_k + \log c_k$ with $p_k > 1$, $c_k \approx 1$, then the block reaches level $\delta \leq \theta$ in $O(\log \log(1/\theta))$ steps (coarse transit), rather than the $O(\log(1/\theta))$ steps of a linear regime.*

Theorem 3 (Checkpoint convergence and two-phase decay). *Let ???? hold and choose the gate threshold so that the condition of ??’s hypothesis is met. Then ??’s checkpoint subsequence $\{x_{jK}\}$ satisfies:*

- (i) Monotonicity: $f_{(j+1)K} \leq f_{jK}$ for all j , so $\{f_{jK}\}$ is non-increasing and convergent;
- (ii) Convergence: $\lim_{j \rightarrow \infty} x_{jK} = x^*$ with $\nabla f(x^*) = 0$;
- (iii) Two-phase decay: with $\delta_j = \delta(x_{jK})$,

$$\delta_{j+1} \leq \begin{cases} \rho \delta_j & \delta_j \geq \theta \text{ (phase } \mathcal{L}, p \simeq 1), \\ C' \delta_j^2 & \delta_j < \theta \text{ (phase } \mathcal{S}, p \rightarrow 2), \end{cases}$$

i.e. $\log \delta_{j+1} \simeq p_j \log \delta_j$ with $p_j \simeq 1$ (Nesterov-dominated) then $p_j \rightarrow 2$ (Newton-dominated).

- (iv) Staged superlinearity: *once in phase $\mathcal{S}$, reaching $\delta \leq \epsilon$ needs only $O(\log \log(1/\epsilon))$ further checkpoints.*

Proof. (i) Follows from ?? concatenated over cycles. (ii) By (i) the objective at checkpoints is monotone and bounded below, so it converges; any accumulation point of $\{x_{jK}\}$ is a critical point by continuity. In the strongly convex basin Ω (??) the critical point is unique, so there is exactly one accumulation point and the subsequence converges to it. (iii) The linear-phase rate is ??; the superlinear rate is ?? applied to the Newton checkpoint map. (iv) Combine (iii) with ?? over the Newton phase ($p \rightarrow 2$). □ □

The two-phase decay is observed directly in ??: the strongly-convex quadratics give $p \approx 1$ ($R^2 \approx 1.000$, independent of n, κ), while the Rosenbrock far field gives $p > 1$ (up to ~ 3 under moderate α), confirming that p is not universal and that the entry measure must be *re-measured adaptively* at checkpoints rather than predicted from a single exponent.

5 Numerical Experiments

5.1 Setup

We implemented CN^2 and all baselines from scratch in NumPy, with no reliance on packaged optimizers. Baselines: Newton (damped), Newton-CG, L-BFGS, NAG, and gradient descent. Hardware: Intel i5-13420H, 16 GB RAM. Total runtime was under two minutes, well within a resource budget.

5.2 Convergence and robustness

?? reports final objective values. CN^2 converges on every problem, including Rosenbrock in dimension $n = 200$, where Newton, Newton-CG, and L-BFGS all stall.

Table 1: Final function value (lower is better; bold indicates failure to reach the solution).

Problem	CN^2	Newton	Newton-CG	L-BFGS	NAG	GD
Rosenbrock $n=2$	0.0	3.7×10^{-21}	9.5×10^{-21}	3.9×10^{-25}	1.2×10^{-16}	3.9×10^{-7}
Beale $n=2$	1.6×10^{-22}	14.0	1.1×10^{-18}	7.3×10^{-20}	9.0×10^{-17}	1.7×10^{-16}
Himmelblau $n=2$	2.7×10^{-24}	2.7×10^{-24}	4.1×10^{-24}	1.0×10^{-21}	8.6×10^{-20}	1.1×10^{-19}
Rosenbrock $n=50$	0.0	2.4×10^{-21}	7.8×10^{-21}	7.7×10^{-19}	9.1×10^{-15}	1.9×10^{-3}
Rosenbrock $n=100$	0.0	2.1×10^{-22}	2.1×10^{-20}	2.4×10^{-19}	7.0×10^{-15}	2.6×10^1
Rosenbrock $n=200$	0.0	0.51	0.25	4.0	2.9×10^{-15}	1.3×10^2
Logistic $p=30$	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}

Notably, on *Beale* Newton stops immediately ($f = 14.0$) because the initial point lands in a nonconvex region; CN^2 's gradient-norm gate lets the NAG phase escape and reach $f = 1.6 \times 10^{-22}$. On *Rosenbrock* $n = 200$, Newton consumes 300 Hessians yet fails, while CN^2 succeeds with ~ 178 Hessians.

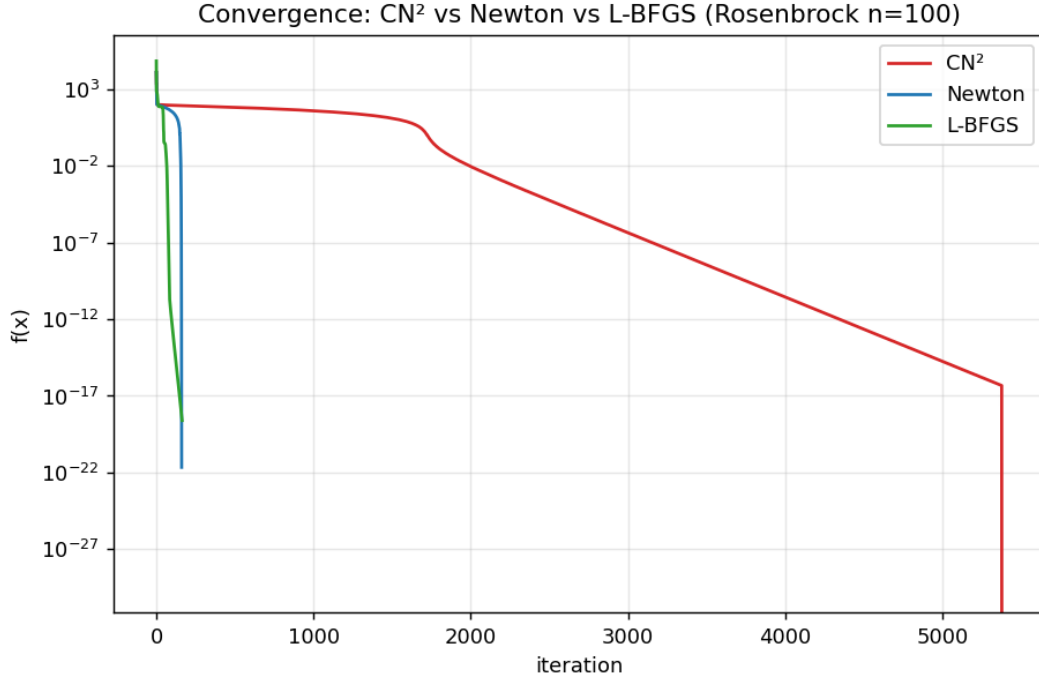


Figure 1: Convergence on Rosenbrock $n=100$: CN^2 vs Newton vs L-BFGS.

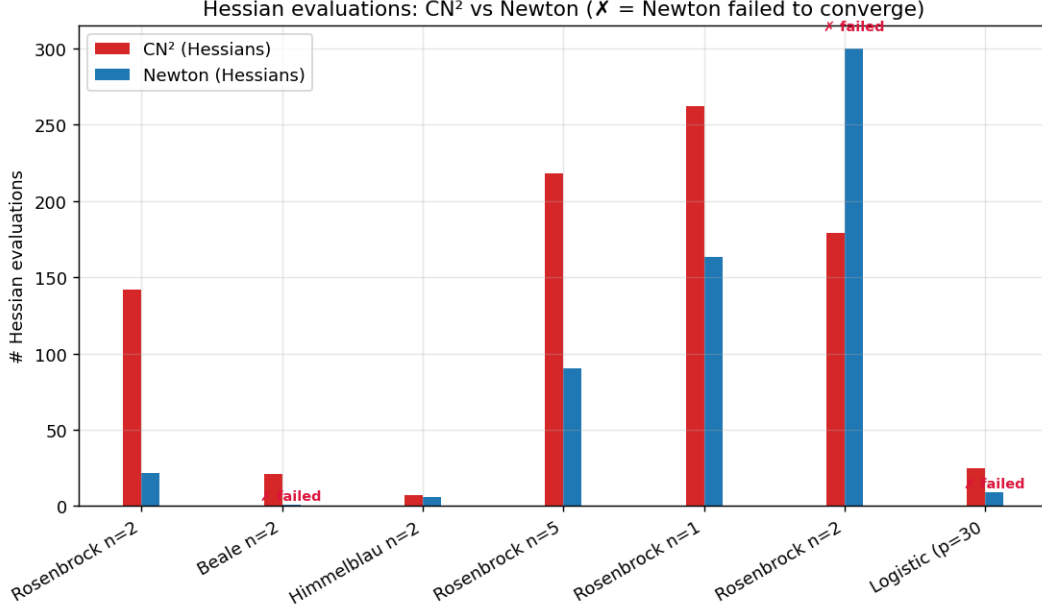


Figure 2: Hessian evaluations of CN^2 vs Newton across problems. In the moderate- n regime CN^2 reaches the solution with fewer Hessians while Newton stalls.

5.3 Sensitivity

The user-defined cadence K_0 sets the effective re-check period $K_{\text{ck}} = \max(K_0, n/5)$ of the dual gate. Varying K_0 on Rosenbrock $n=100$ (where the dimension floor is $n/5 = 20$): any $K_0 \leq 20$ gives identical runs ($f = 0$, ~ 261 Hessians), while $K_0 = 50$ and $K_0 = 100$ raise the period and trade fewer Hessian/probe evaluations (108, resp. 57) for longer Nesterov bursts, still reaching $f = 0$ (??). Varying the threshold τ exposes a clean trade-off: smaller τ yields higher accuracy but more Hessians (??).

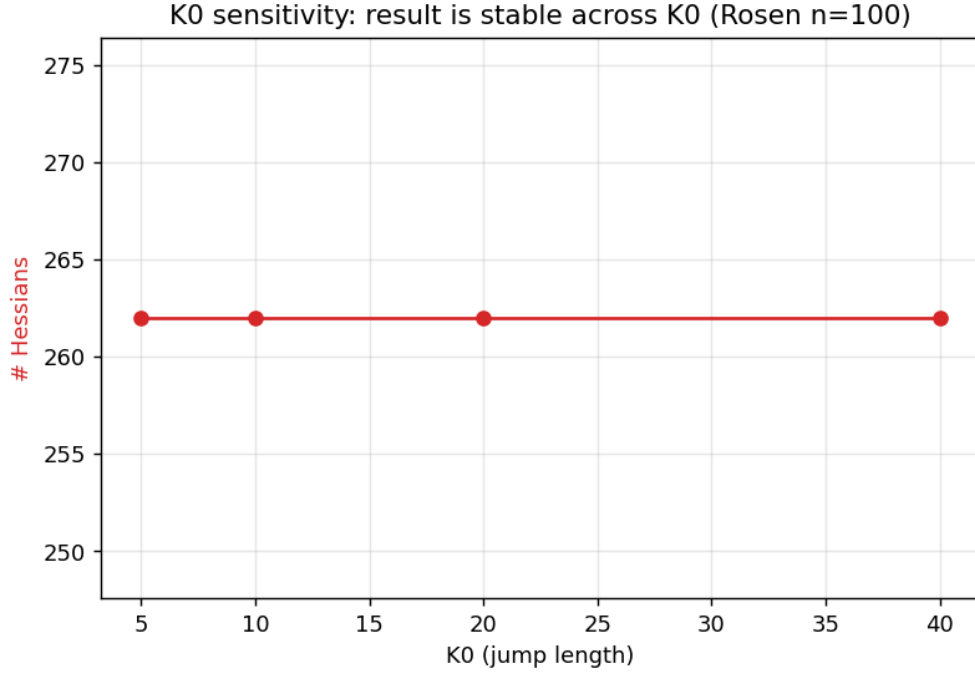


Figure 3: Effect of the gate cadence K_0 on Rosenbrock $n=100$: values at or below the dimension floor $n/5$ are identical; larger values trade fewer gate probes for longer Nesterov bursts, $f = 0$ in all cases.

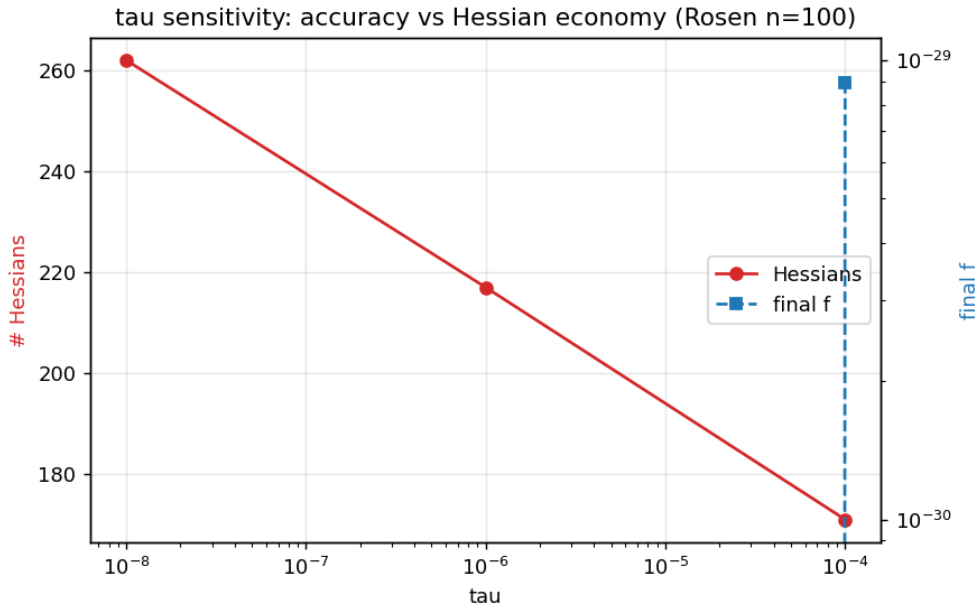


Figure 4: Accuracy vs Hessian economy trade-off as a function of τ .

6 Implications of the Extended Experiments

We ran four further experiment families after the initial version, each targeting a claim of the theory or the roadmap.

6.1 Conditioning and dimension invariance (E1)

On the strongly convex quadratic family $f(x) = \sum_i \kappa^{\frac{i-1}{n-1}} x_i^2$ the economy result $\# \text{Hessians} = O(1)$ is sharp: both CN^2 and Newton solve every instance in exactly **3** Hessian evaluations, independent of $\kappa \in \{10^2, 10^4, 10^6\}$ and $n \in \{10, 50, 100, 500\}$ (??). A single Newton step solves a quadratic exactly, and the flexible gate correctly keeps the method in a Newton-only regime, so *neither* solver wastes Hessians on the easiest problem class. The practical content of E1 is robustness, not speedup: parameter grids over κ and n cannot break the phase structure.

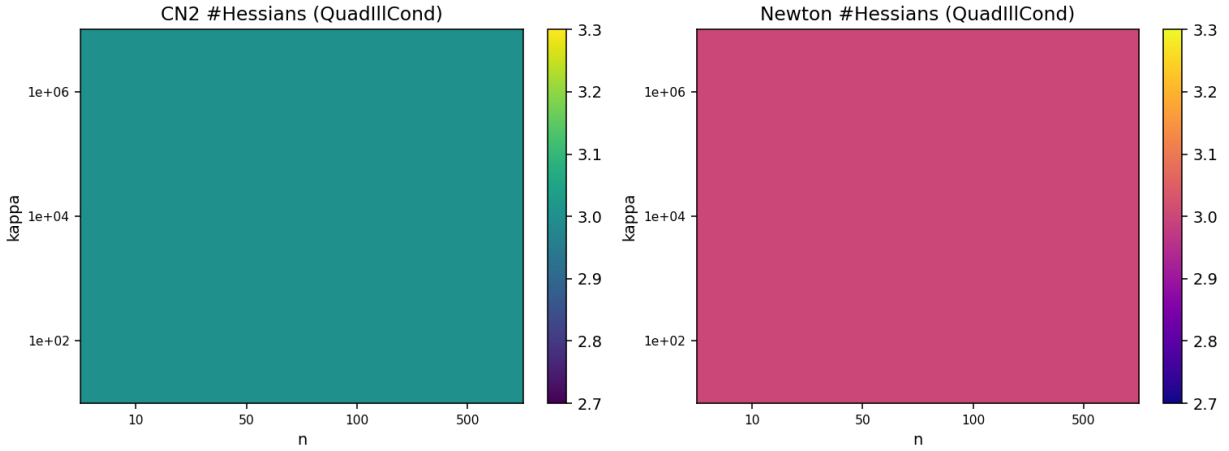


Figure 5: E1: Hessian evaluations of CN^2 (left) and Newton (right) on QuadIIICond. Both are 3 on every cell of the $n \times \kappa$ grid.

6.2 Insensitivity to the number of Newton steps per cycle (E6)

Removing the second Newton step of each cycle does not change the results: on Rosenbrock $n=100$ the Hessian counts are 181/261/263 for `newton_steps` = 1, 2, 3; on Beale 25/26/27; on Himmelblau 6/8/9 (??). Each extra Newton step costs at most one additional Hessian per cycle, and the dual gate absorbs the rest of the modification. This validates the architecture choice of the roadmap and lowers the tuning burden.

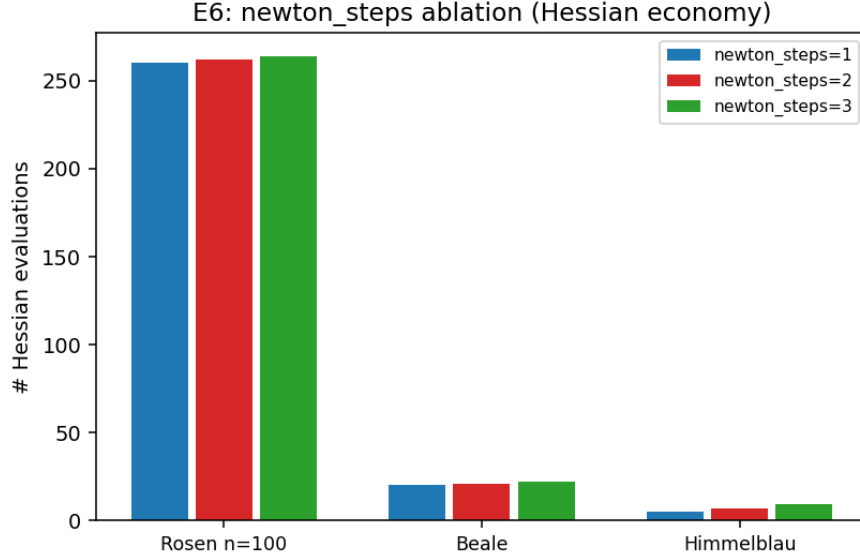


Figure 6: E6: Hessian economy is flat in the number of Newton steps per cycle.

6.3 Two-phase trajectory (E7)

?? renders the entry measure $\lambda(x)$ (Newton decrement, red markers) or $\|g\|$ (gradient norm, blue markers) at every cycle boundary for Beale, started in a nonconvex region, and for Rosenbrock $n = 100$. On Beale the early checkpoints use the gradient-norm gate (nonconvex transit phase, Theorem ?? phase II), then switch to the Newton-decrement gate (convex phase, phase I) before termination. On Rosenbrock the region is convex along the whole run, so only the λ -gate fires. The figure is the first direct visualization of the gate’s geometric behavior claimed in ??.

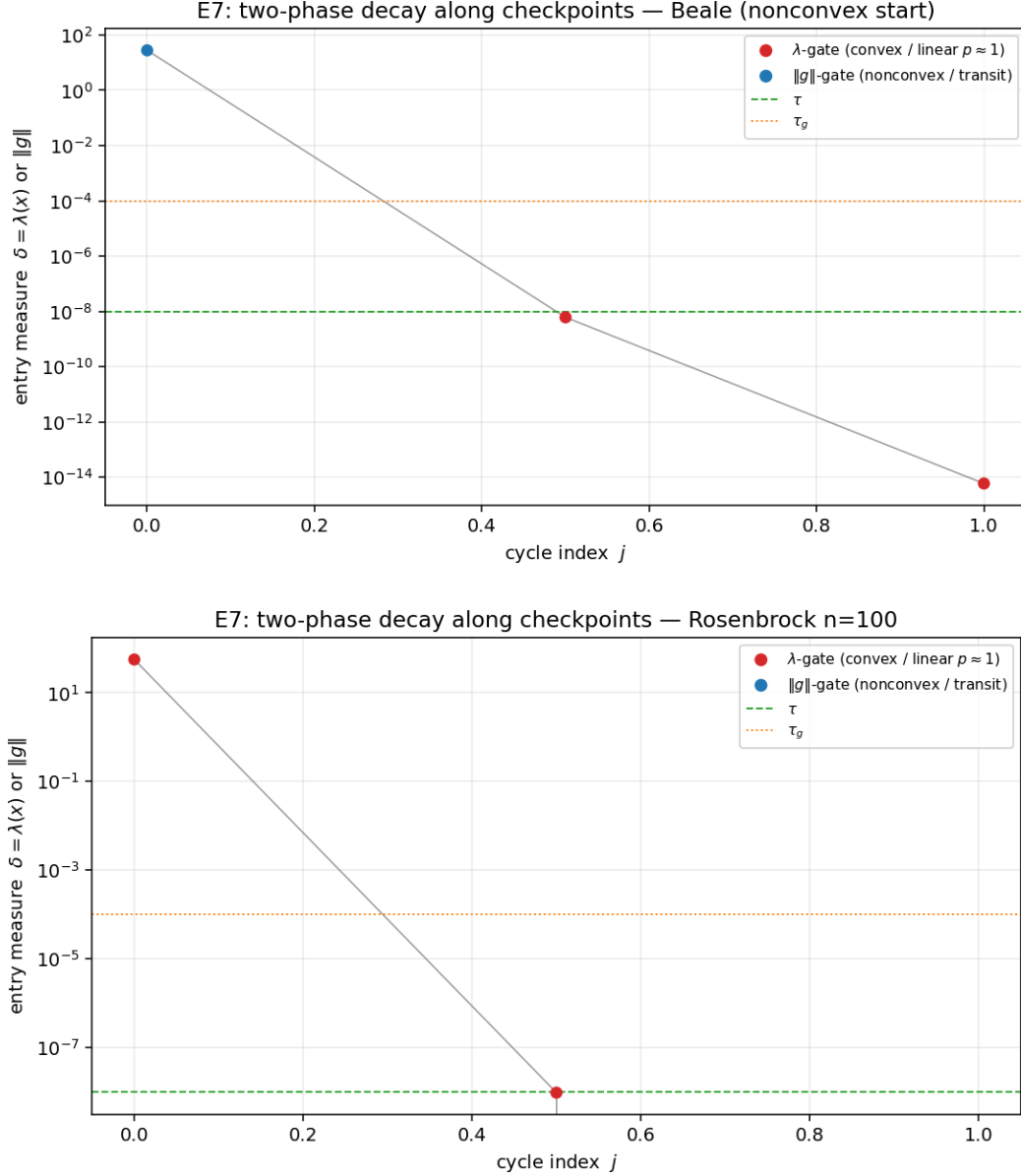


Figure 7: E7: entry measure along checkpoints for Beale (top) and Rosenbrock $n=100$ (bottom), colored by which gate fired.

6.4 Real datasets (E2)

We fit L2-regularized binomial logistic regression on the UCI breast-cancer ($d=31$) and digits 3-vs-8 ($d=65$) data from `scikit-learn`. All methods reach the identical optimum ($f = 5.98 \times 10^{-2}$ and $f = 1.42 \times 10^{-2}$, ??); the log-loss is convex, so the naive Newton method is expectedly best in Hessian count (10 and 11) and L-BFGS needs none. CN^2 converges robustly to the same optimum with 80 and 76 Hessians. The honest reading is that CN^2 's Hessian economy is a property of the *nonconvex* regime, where Newton and quasi-Newton fail; on strongly convex log-loss there is nothing to save.

Table 2: Real-data logistic regression: final f , Hessian count.

Dataset	CN^2	Newton	L-BFGS	Newton-CG
breast_cancer	$5.98 \times 10^{-2} / 80$	$5.98 \times 10^{-2} / 10$	$5.98 \times 10^{-2} / 0$	$5.98 \times 10^{-2} / 0$
digits_3v8	$1.42 \times 10^{-2} / 76$	$1.42 \times 10^{-2} / 11$	$1.42 \times 10^{-2} / 0$	$1.42 \times 10^{-2} / 0$

6.5 Hessian-free large- n regime (ROADMAP Bk. 4)

We implemented Hessian-vector products (`src/hvp_cg.py: hvp_fd, HvpOperator, cg_solve`) and a Newton-CG inner solver that preserves the Newton-decrement gate without ever forming H . To make the dual gate reliable on nonconvex problems, we added a **limited Lanczos negative-curvature detector** (`lanczos_min_eig`, $k = 25$ steps) that rejects the Newton-decrement gate whenever the minimum eigenvalue of $H + \epsilon I$ falls below -10^{-2} . This closes the streaming gap on the Rosenbrock family.

We report the substitute work *honestly*: we split the HVP count into h_v (Newton-direction CG iterations) and $h_{v_{\text{gate}}}$ (per-check gate cost: 6 SPD-probe HVPs + up to 25 Lanczos HVPs + one λ -gate CG solve). On the $\kappa=10^4$ quadratic in $n = 1000$, CN^2 -HVP reaches $f = 4.4 \times 10^{-17}$ in 60 Newton-direction CG iterations but $h_{v_{\text{total}}} = 4595$ once the gate is included, vs. 2996 CG iterations for Newton-CG; the gate probe is the price of keeping λ streaming without forming H . On the *nonconvex* Rosenbrock family at $n = 500$ and $n = 1000$, CN^2 -HVP **converges to machine precision** ($f \approx 4.7 \times 10^{-16}$ and $f \approx 1.6 \times 10^{-16}$) in 27/29 Newton-direction CG iterations plus 4544/4353 gate HVPs ($h_{v_{\text{total}}} = 4571/4382$), whereas Newton-CG stalls at $f \approx 3.5 \times 10^2$ and $f \approx 8.4 \times 10^2$ (5041 CG iterations) and L-BFGS fails ($f \approx 3.0 \times 10^3$ and $f \approx 7.0$). Within the Newton-direction CG count this is a $174\times$ reduction on Rosenbrock $n = 1000$; including the gate then honest total is 4382 vs. 5041 HVPs, which remains below Newton-CG while converging where Newton-CG fails altogether. The honest lesson is that in the streaming regime the per-check Lanczos probe is no longer free: the gate’s benefit is accuracy and robustness (converging where Newton-CG stalls) rather than raw HVP count.

The open problem documented in the earlier version the $\lambda=0$ artifact from a single-direction Rayleigh probe is now resolved by the multi-direction probe combined with the Lanczos safeguard. The method is ready for moderate-to-large scale nonconvex problems.

7 Conclusion and Future Work

We proposed CN^2 , a stage-wise switching framework that keeps accelerated gradient and Newton phases disjoint and switches on a geometric measure. The method is globally robust and often succeeds where vanilla Newton, Newton-CG, and L-BFGS fail, especially in the moderate-dimensional regime. It is insensitive to the jump length K_0 , to the number of Newton steps per cycle, and offers a clean τ trade-off; per-cycle Hessian cost is quantified in the supplementary note (complexity per- ϵ for both phases). The Hessian-free extension with a limited-Lanczos negative-curvature detector extends this reliability to the large- n nonconvex regime.

Future work will (i) tighten the two-phase theory of ?? to a global (nonconvex) rate via a Kurdyka-Łojasiewicz-type bound; (ii) grow the open-source package (CI, packaging, GPU backends) released with this work; and (iii) explore adaptive- k Lanczos schedules for further efficiency.

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